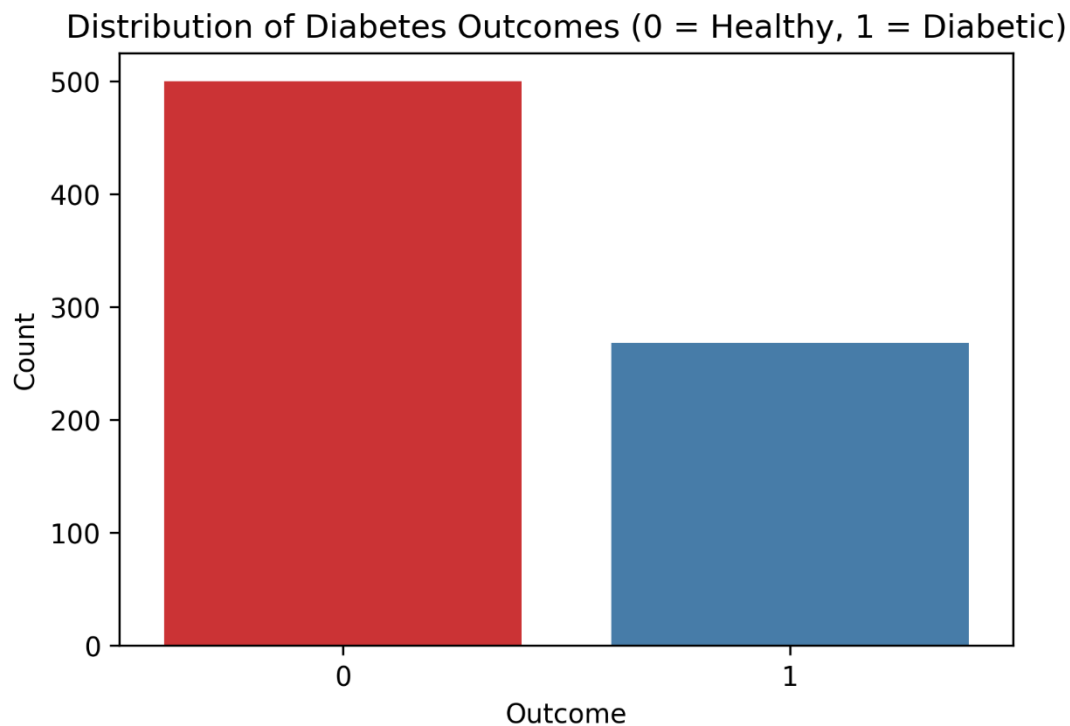


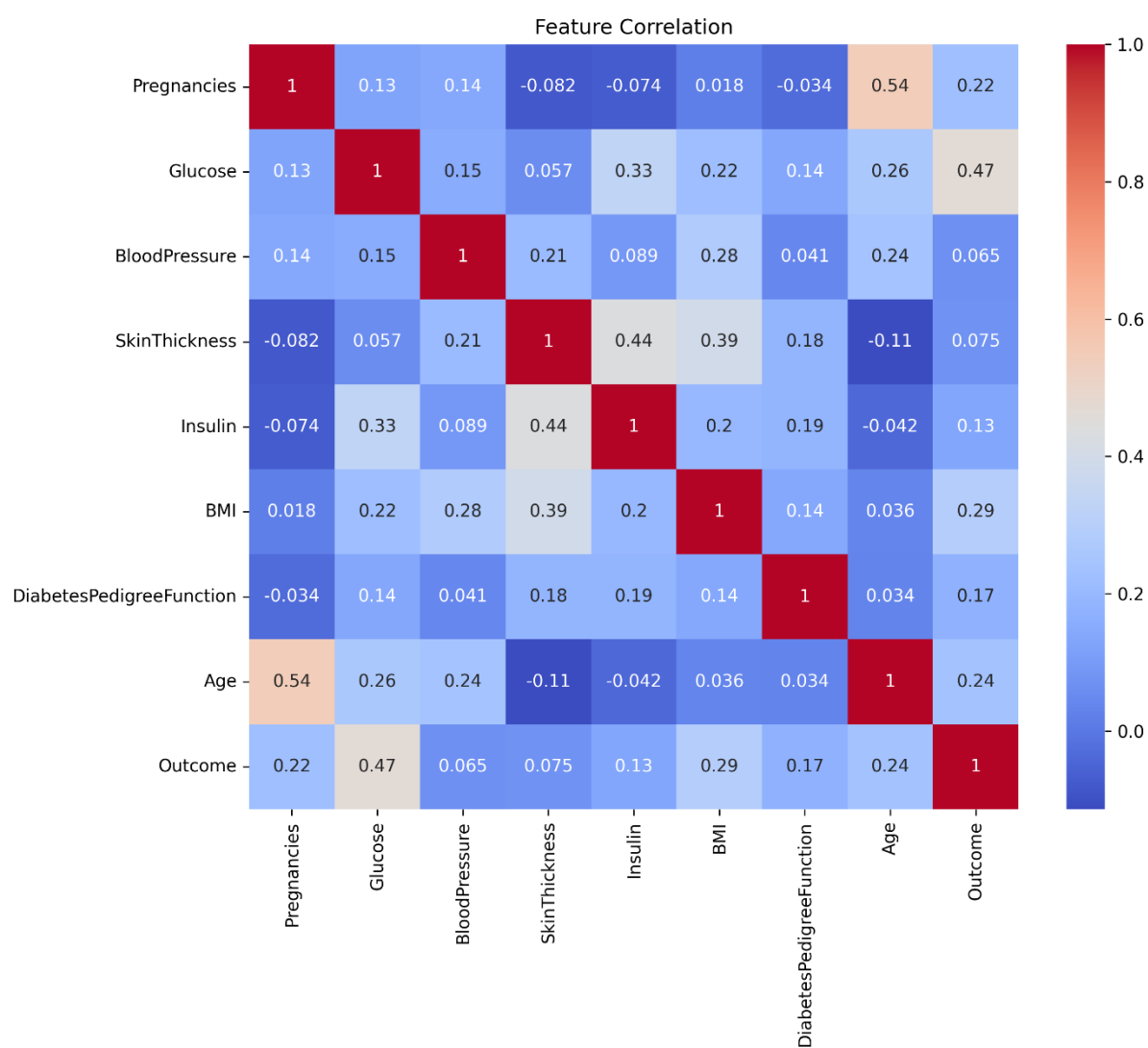
Graphs

Graph 1: Target Class Distribution:



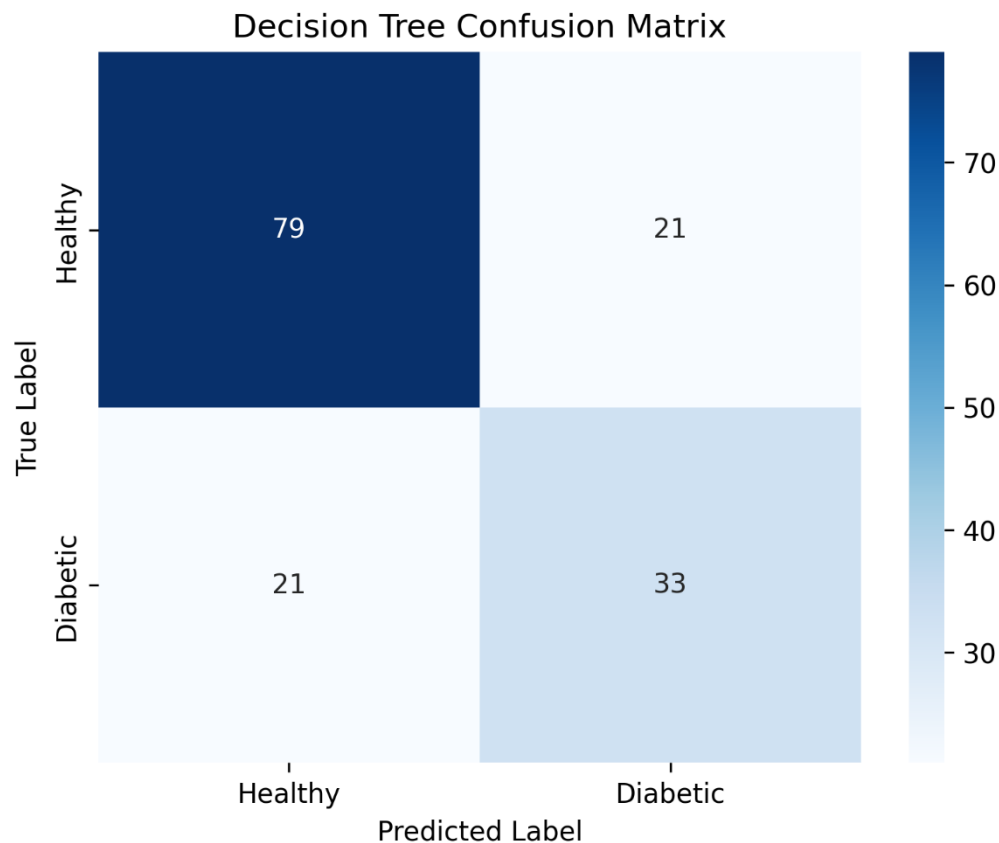
This chart shows the distribution of healthy (0) and diabetic (1) patients in the dataset. The dataset is moderately imbalanced, with healthy cases occurring more frequently than diabetic cases. This imbalance motivates the use of `class_weight="balanced"` during model training.

Graph 2: Correlation Heatmap



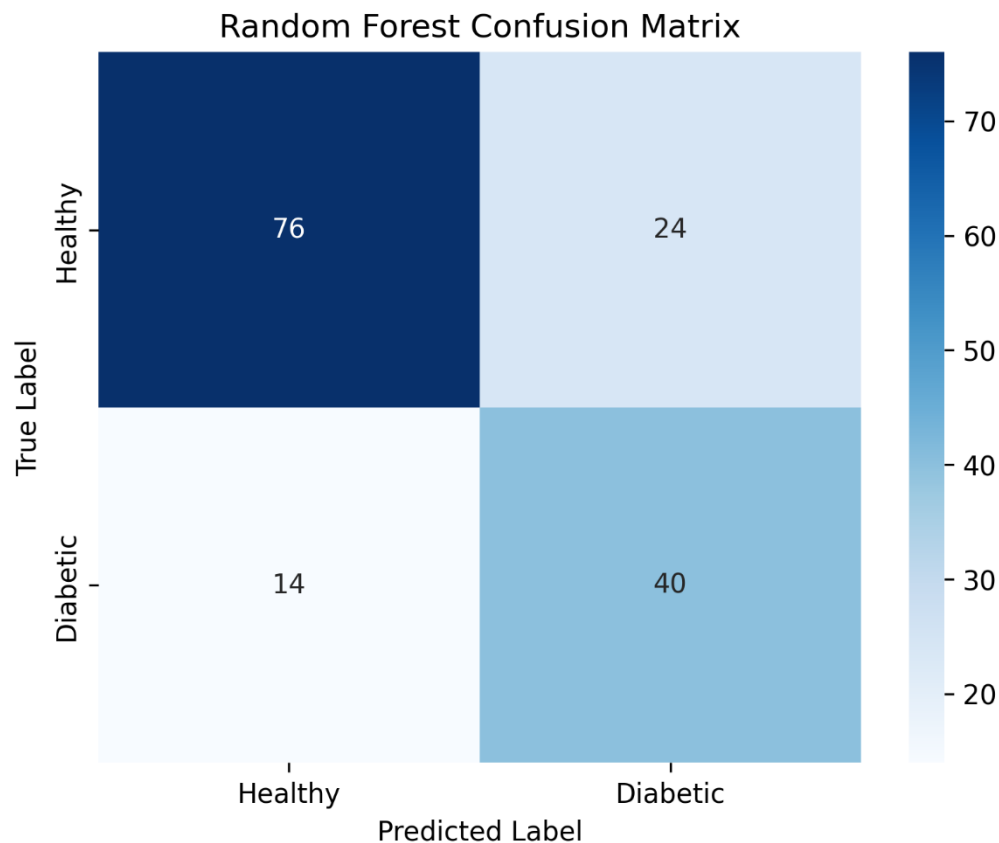
The heatmap visualizes the correlation between features. It helps identify relationships among medical measurements and the target variable. Features such as Glucose generally show a stronger correlation with diabetes outcome than others.

Graph 3: Decision Tree Confusion Matrix:



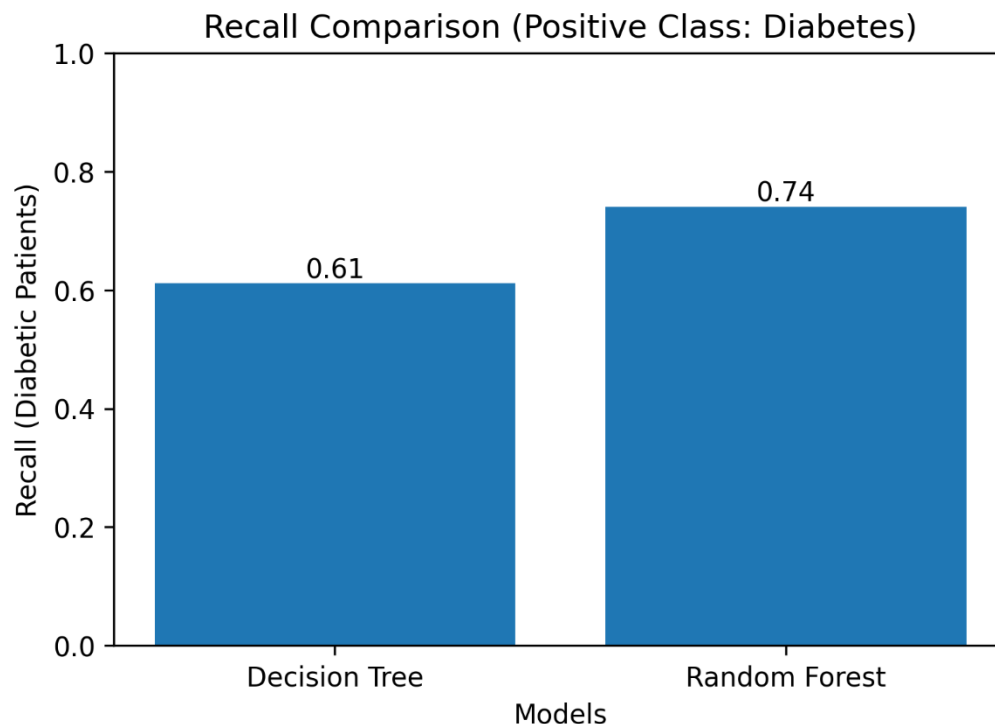
The confusion matrix summarizes the Decision Tree model's predictions. It shows correctly classified and misclassified healthy and diabetic patients, helping identify false positives and false negatives.

Graph 4: Random Forest Confusion Matrix:



The Random Forest confusion matrix demonstrates improved identification of diabetic patients after using `class_weight="balanced"`, while maintaining competitive overall performance.

Graph 5: Recall Comparison:



This chart compares the recall for the diabetic class (Outcome = 1). Recall measures the proportion of actual diabetic patients correctly identified by the model. The Random Forest model achieved a recall of **0.74**, outperforming the Decision Tree (**0.61**). Since missing diabetic patients can delay diagnosis and treatment, higher recall is particularly important for healthcare applications.